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**IEEE Chandigarh Subsection International Conference : Submission (1227) has been created.**

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Thu, Apr 30, 2026 at 12:56 PM

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The following submission has been created.

Track Name: Next-Generation Computing and Applications

Paper ID: 1227

Paper Title: Neuromorphic Computing Chips for Low-Power Edge AI Applications

**Abstract:**

Small portable devices such as smartwatches, security cameras, and industrial sensors are increasingly using modern artificial intelligence (AI) systems. However, these devices are not able to use traditional computer chips due to the high power consumption of those processors. Neuromorphic computing offers a more efficient alternative, and thus creates chips that mimic how the brain processes information. Instead of processing all data continuously, these chips will only become active when there is something meaningful to process. Because of this, they significantly decrease an AI processor's energy consumption. This study examines the functioning of neuromorphic chips, their necessity, and performance when compared to conventional chips. We focus on two major systems (Intel Loihi and IBM TrueNorth) and new alternative research systems. The results of our study demonstrate that neuromorphic chips can reduce energy consumption by 90–96% versus standard GPUs while achieving accuracy of 89–97%, and process information in less than 3 milliseconds (i.e. fast enough for real-time tasks). Consideration has also been given to those software tools and hardware platforms utilized to build and test neuromorphic processor architectures. The findings confirm that neuromorphic architectures are now ready for application in areas such as health care, robotics, smart cities and industrial monitoring.

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**Submission Questions Response:**

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4. Co-author's Consent  
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7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

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